

Statistics

- Q1** A class consists of 50 students, out of which 30 are girls. The mean of marks scored by girls in a test is 73 (out of 100), and that of boys is 71. Determine the mean score of the whole class.
- Q2** Find the median for the data 8, 5, 7, 10, 15, 21.
- Q3** For a moderately skewed distribution, mean = 12 and mode = 6. Using these values, find the value of the median.
- Q4** The marks scored by a student in different subjects are 45, 91, 62, 71, 55. Find the median of the given data using the median formula.
- Q5** For any given data, the mean is 45.5, and the median is 43. Find the modal value.
- Q6** Let X be random variable with mean μ_x and variance $\sigma_x^2 > 0$, then $\text{Var}(aX + b)$ is
 (A) $a\sigma_x^2$ (B) $a^2\sigma_x^2$
 (C) $a\sigma_x^2 + b$ (D) $a^2\sigma_x^2 + b$
- Q7** If X and Y are two random variables with respective expected values $E(X)$ and $E(Y)$, then $\text{Cov}(X, Y)$ is-
 (A) $E(XY) - E(X)E(Y)$
 (B) $E(X/Y) = E(XY)$
 (C) $E(XY) - E(X/Y)$
 (D) $E(XY) - E(X)/E(Y)$
- Q8** The covariance of two independent variate is equal to-
 (A) zero
 (B) unity
 (C) sum of their expectation
 (D) The product of their expectations
- Q9** If X_1 and X_2 are independent, then $V(X_1 - X_2)$ is equal to :
- (A) $V(X_1) + V(X_2)$
 (B) $V(X_1) - V(X_2) - 2 \text{Cov}(X_1, X_2)$
 (C) $V(X_1) - V(X_2)$
 (D) $V(X_1) + V(X_2) - 2 \text{Cov}(X_1, X_2)$
- Q10** A random variable X has the following probability function:
- | | | | | | | | |
|--------|-----|------|------|------|------|-------|-------|
| x | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
| $f(x)$ | k | $3k$ | $5k$ | $7k$ | $9k$ | $11k$ | $13k$ |
- (i) Find k ,
 (ii) Find $P(\geq 5)$, $P(3 < X \leq 6)$, $P(X < 4)$
- Q11** If X is the number of points rolled with a balanced die, find the expected value of $g(X) = 2X^2 + 1$
- Q12** A discrete random variable X has the following probability distribution:
- | | | | | | |
|------------|------|------|------|-----|------|
| X | 1 | 2 | 3 | 4 | 5 |
| $P(X = x)$ | 0.05 | 0.15 | 0.35 | 0.4 | 0.05 |
- Calculate:**
 (i) $F(3)$
 (ii) $E[X]$
 (iii) $\text{var}[X]$
- Q13** If random variable X assumes only positive integral values, with the probability
 $P(X = x) = \frac{2}{3} \left(\frac{1}{3}\right)^{x-1}$, $x = 1, 2, 3, \dots$, then $E(X)$ is
 (A) $2/9$ (B) $2/3$
 (C) 1 (D) $3/2$
- Q14** A discrete random variable X has the following probability function
- | | | | | | |
|-----|-----|-----|-----|-----|-----|
| x | 10 | 20 | 30 | 40 | 50 |
| y | 0.1 | 0.1 | 0.4 | 0.3 | 0.1 |
- Denote by μ_x and σ_x the mean and the standard deviation of X . Find $P(|X - \mu_x| \leq \sigma_x)$



- (A) 1 (B) 0.8
(C) 0.7 (D) 0.5
- Q15** An unbiased coin is tossed until a head is obtained. If N denotes the number of tosses required, what is $P(N > 1)$?
(A) $1/2$ (B) 1
(C) $1/4$ (D) $1/8$
- Q16** Probability mass function of a discrete random variables are given below.
 $X_i = 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6$
 $P(X_i) = k \quad 3k \quad 5k \quad 7k \quad 9k \quad 11k \quad 13k$
 (A) Find K
 (B) Evaluate $P(X < 4)$
 (C) Evaluate $P(3 < x \leq 6)$
- Q17** Which one of the following is not possible for a binomial distribution?
 (A) Mean = 2, variance = $3/2$
 (B) Mean = 5, variance = 9
 (C) Mean = 10, variance = 5
 (D) Mean = 4, variance = $8/3$
- Q18** A coin is twice as likely to turn up tails as heads. If the coin is tossed independently, what is the probability that the third head occurs on the fifth toss?
 (A) $\frac{8}{81}$ (B) $\frac{40}{243}$
 (C) $\frac{16}{81}$ (D) $\frac{80}{243}$
- Q19** If X follows binomial distribution with parameters n and p , the variance of X/n is-
 (A) $\frac{p(1-p)}{n}$ (B) $np(1-p)$
 (C) $p(1-p)$ (D) $\frac{p(1-p)}{n^2}$
- Q20** In a class of Pathology, the professor decided to conduct a test of swine flu on all the students. Test result says that one student in every ten is having swine flu. What is the probability that out of 5 students expected to attend a class, at least 4 will not have swine flu?
- Q21** If the probability of a defective item is 0.1, find (i) the mean (ii) standard deviation for the

distribution of items in a total of 400.

- Q22** The mean and variance of a binomial distribution are 2.5 and 1.875 respectively. Obtain the binomial probability distribution.
- Q23** 100 dice are thrown. How many are expected to fall 6. What is the variance in the number of 6's?
- Q24** Let X be a poisson random variable and $P(X = 1) + 2 P(X = 0) = 12 P(X = 2)$. Which one of the following statements is TRUE?
 (A) $0.40 < P(X = 0) \leq 0.45$
 (B) $0.40 < P(X = 0) \leq 0.50$
 (C) $0.50 < P(X = 0) \leq 0.55$
 (D) $0.55 < P(X = 0) \leq 0.60$
- Q25** In a Poisson distribution, the probability of observing 3 is $2/3$ times that of observing 4. The mean of the distribution is
 (A) 5 (B) 6
 (C) 7 (D) 8
- Q26** The number of misprints per page of a book (X) follows the poisson distribution such that $P(X = 1) = P(X = 2)$. If the book contains 500 pages, the expected number of pages containing at most one misprint is-
 (A) $5005e^{-2}$
 (B) $10005e^{-2}$
 (C) $15005e^{-2}$
 (D) $500(1 - 3e^{-2})$
- Q27** A manufacturing company supplies condensers with 1% defective pieces. Condensers are packed in boxes of 100. Find the probability that a box picked at random will have four or more faulty condensers.
- Q28** One fifth percent of the blades produced by a manufacturing factory turn out to be defective. The blades are supplied in packets of 10. Use Poisson distribution to calculate the approximate number of packets containing no defective, one defective and two defective



blades respectively in a consignment of 1,00,000 packets. Given $e^{-0.02} = 0.9802$

Q29 A certain company sells tractors which fail at a rate of 1 out of 1000. If 500 tractors are

purchased from this company what is the probability of 2 of them failing within first year?



Answer Key

Q1 72.2
Q2 9
Q3 10
Q4 62
Q5 38
Q6 (A)
Q7 (A)
Q8 (A)
Q9 (D)
Q10 16/49.
Q11 ?
Q12 = 0.88
Q13 (A)
Q14 (A)
Q15 (A)

Q16 33/49
Q17 (A)
Q18 (A)
Q19 (A)
Q20 $\left(\frac{9}{10}\right)^4 \times \frac{7}{5}$
Q21 6
Q22 10
Q23 = 13.9
Q24 (A)
Q25 (A)
Q26 (A)
Q27 0.019
Q28 $19.6 \cong 20.$
Q29 0.07582



Hints & Solutions

Q1 Text Solution:

Given,

Total number of students in a class = 50

Number of girls in the class = 30

Number of boys in the class = $50 - 30 = 20$

Mean marks scored by girls = 73

Mean marks scored by boys = 71

Thus, the total marks scored by girls = $73 \times 30 = 2190$

Also, the total marks scored by boys = $71 \times 20 = 1420$

Mean score of the class = (Total marks scored by girls and boys)/Total number of students

$$\begin{aligned} &= (2190 + 1420)/50 \\ &= 3610/50 \\ &= 72.2 \end{aligned}$$

Q2 Text Solution:

Arranging the given data in ascending order, we get, 5, 7, 8, 10, 15, 21.

Here, the number of observations is 6, which is even.

Hence, Median = $[(n/2)^{\text{th}} \text{ term} + ((n/2) + 1)^{\text{th}} \text{ term}]/2$

Median = $[(6/2)^{\text{th}} \text{ term} + ((6/2) + 1)^{\text{th}} \text{ term}]/2$

Median = $(3^{\text{rd}} \text{ term} + 4^{\text{th}} \text{ term})/2$

Here, 3^{rd} term = 8 and 4^{th} term = 10

Therefore, median = $(8+10)/2 = 18/2 = 9$

Hence, the median of the given data is 9.

Q3 Text Solution:

Given that, mean = 12 and mode = 6

We know that, $3 \text{ Median} = 2 \text{ Mean} + \text{Mode}$

Now, substitute the values in the formula, we get

$$3 \text{ Median} = 2(12) + 6$$

$$3 \text{ Median} = 24 + 6$$

$$3 \text{ Median} = 30$$

$$\text{Median} = 30/3 = 10.$$

Hence, the value of median is 10.

Q4 Text Solution:

Given data: 45, 91, 62, 71, 55

Arranging the given set in ascending order, we get 45, 55, 62, 71, 91.

Here, we have an odd number of observations, (i.e.) 5.

If "n" is odd, the median formula is $[(n+1)/2]^{\text{th}}$ term

$$= [(5+1)/2]^{\text{th}} \text{ term}$$

$$= [(6)/2]^{\text{th}} \text{ term} = 3^{\text{rd}} \text{ term} = 62.$$

Hence, the median of the given data is 62.

Q5 Text Solution:

We know that,

$$\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$$

$$\therefore \text{Mode} = 3 \times 43 - 2 \times 45.5$$

$$= 129 - 91 = 38.$$

$$\text{Mode} = 38.$$

Q6 Text Solution:

Mean = μ_x = variance is $(6x)^2 > 0$

$$\text{Var}(ax + b) = \text{Var}(ax) + \text{Var}(b)$$

$$= a^2 \text{Var}(x)$$

$$= a^2 \cdot 6x^2$$

Q9 Text Solution:

$$\text{Var}(ax + by) = a^2 \text{Var}(x) + b^2 \text{Var}(y) + 2ab \text{Cov}(x, y)$$

$$\Rightarrow \text{Var}(x - y) = 1 \cdot \text{Var}(x) + 1 \cdot \text{Var}(y) + 0$$

Q10 Text Solution:

$$(i) \sum f(x) = 1$$

$$K + 3K + 5K + 7K + 9K + 11K + 13K = 1$$

$$K \left(\overset{1}{1} + \overset{3}{3} + \overset{5}{5} + \overset{7}{7} + \overset{9}{9} + 11 + 13 \right) = 1$$

$$K(10 + 10 + 29) = 1$$

$$K \times 49 = 1$$

$$K = \frac{1}{49}$$

$$(ii) P(X \geq 5) = P(X = 5) + P(X = 6)$$

$$= 11K + 13K = 24K = 24/49$$

$$P(3 < X \leq 6) = P(X = 4) + P(X = 5) + P(X = 6)$$

$$= 9K + 11K + 13K$$

$$= 20K + 13K$$

$$= 33K = 33/49$$

$$P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$



$$= K + 3K + 5K + 7K$$

$$= 16K = 16/49.$$

Q11 Text Solution:

$$g(X) = 2X^2 + 1$$

X =	0	1	2	3	4	5	6
P(X) =	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$g(X) = 2X^2 + 1$$

$$E(g(x)) = E(2X^2 + 1)$$

$$= E(2X^2) + E(1)$$

$$= 2E(X^2) + (1)$$

$$= 2 \left[\sum x_i^2 P(x_i) \right] + 1$$

$$= 2$$

$$\left[\left(0 \times \frac{1}{6}\right) + \left(1 \times \frac{1}{6}\right) + \left(4 \times \frac{1}{6}\right) + \left(9 \times \frac{1}{6}\right) \right.$$

$$\left. + \left(16 \times \frac{1}{6}\right) + \left(25 \times \frac{1}{6}\right) + \left(36 \times \frac{1}{6}\right) \right] + 1$$

$$= 2 \left[\left(\frac{1}{6}\right) (0 + 1 + 4 + 9 + 16 + 25 + 36) \right]$$

$$+ 1$$

$$= 27 + 1$$

$$= 28.$$

Q12 Text Solution:

$$(i) f(3) = f(x) = P(X \leq n)$$

$$= P(X \leq 3)$$

$$= P(X = 1) + P(X = 2) + P(X = 3)$$

$$= 0.05 + 0.15 + 0.35$$

$$= 0.05 + 0.50 = 0.55$$

$$(ii) E(x) = \sum x_i P(x_i)$$

$$=$$

$$1 \times 0.05 + 2 \times 0.15 + 3 \times 0.35 + 4 \times 0.40 + 5 \times 0.05$$

$$= 3.25$$

$$(iii). V(x) = E(x^2) - E(x)^2$$

$$\Rightarrow E(x^2) = \sum (x_i)^2 P(x_i)$$

$$= 0.05 + 0.6 + 3.15 + 6.4 + 1.25$$

$$= 11.45$$

$$V(X) = 11.45 - (3.25)^2$$

$$= 0.88$$

Q13 Text Solution:

$$E(X) = \sum_{i=1}^x x_i P(x_i)$$

$$= 1 \times \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^{1-1} + 2 \times \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^{2-1} + 3$$

$$\times \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^{3-1}$$

$$= \frac{2}{3}$$

$$\left[1 \times \left(\frac{1}{3}\right)^0 + 2 \times \left(\frac{1}{3}\right)^1 + 3 \times \left(\frac{1}{3}\right)^2 + 4 \right.$$

$$\left. \times \left(\frac{1}{3}\right)^3 \right]$$

$$= \frac{2}{3} \left[\sum_{x=1}^x x \left(\frac{1}{3}\right)^{n-1} \right]$$

$$S = \left(\frac{1}{3}\right)^0 + 2 \times \left(\frac{1}{3}\right)^1 + 3 \times \left(\frac{1}{3}\right)^2$$

Q14 Text Solution:

$$P(|X - \mu_x| \leq 6_x)$$

$$\text{Here Mean} = \sum x_i P(x_i)$$

$$= 10 \times 0.1 + 20 \times 0.1 + 30 \times 0.4 +$$

$$40 \times 0.3 + 50 \times 0.1$$

$$= 1 + 2 + 12 + 12 + 5$$

$$= 24 + 8 = 32.$$

$$\text{Variance} = E(X^2) - E(X)^2$$

$$= E(X^2) = \sum (x_i)^2 P(x_i)$$

$$= 100 \times 0.1 + 400 \times 0.1 + 900 \times 0.4$$

$$+ 1600 \times 0.3 + 2500 \times 0.1$$

$$= 1140$$

$$\text{Var}(x) = 1140 - 1024 = 116$$

$$= P(|X - 32| \leq 116)$$

$$= 116 \leq x - 32 \leq 116$$

$$= -84 \leq x \leq 148$$

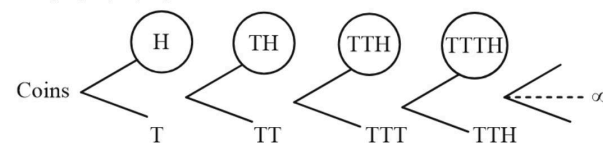
$$P(0 \leq x \leq 148)$$

$$0.1 + 0.1 + 0.4 + 0.3 + 0.1 = 1$$

Q15 Text Solution:

$$N = \{\text{Number of tosses required}\}$$

$$= \{1, 2, 3, 4, 5, \dots\}$$



$$S = \{H, TH, TTH, TTTH, \dots\}$$

N:	1	2	3	4	5	6
P(N)	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$

$$P(N > 1) = 1 - P(N = 1) = 1 - \frac{1}{2} = \frac{1}{2} \text{ i.e.}$$

Q16 Text Solution:

(A) We know that sum of PMF is one.

$$\sum_i P(x_i) = 1$$



$$k + 3k + 5k + 7k + 9k + 11k + 13k = 1$$

$$49k = 1$$

$$k = \frac{1}{49}$$

$$(B) \quad x < 4 \Rightarrow x = 0 \text{ or } x = 1 \text{ or } x = 2 \text{ or } x = 3$$

$$\text{Therefore, } P[x < 4] = p[x = 0] + p(x = 1) + p(x = 2) + p(x = 3) = k + 3k + 5k + 7k = \frac{16}{49}$$

$$(C) \quad 3 < x \leq 6 - x = 4 \text{ or } x = 5 \text{ or } x = 6$$

$$\text{Therefore, } P[3 < x \leq 6] = 9k + 11k + 13k = 33k = \frac{33}{49}$$

Q17 Text Solution:

$$\text{Mean} = np$$

$$\text{Variance} = npq = np(1 - p)$$

$$(A) \text{ Mean} = 2$$

$$\text{Variance} = 3/2$$

$$\frac{\text{Variance}}{\text{Mean}} = \frac{3}{4} = q \quad P = \frac{1}{4}$$

$$np = 2$$

$$n \times \frac{1}{4} = 2$$

$$n = 8, \text{ thus possible.}$$

$$(B) \text{ Mean} = 5$$

$$\text{Variance} = 9$$

$$q = 9/5$$

$$P = 1 - \frac{9}{5} = -\frac{4}{5} \rightarrow \text{Not possible.}$$

$$(c) \text{ Mean} = 10.$$

$$\text{Variance} = 5$$

$$q = \frac{5}{10} = \frac{1}{2}$$

$$P = 1 - \frac{1}{2} = \frac{1}{2}$$

$$n \times \frac{1}{2} = 10$$

$$n = 20 \text{ Possible.}$$

$$(D) np = 4$$

$$npq = 8/3$$

$$q = 2/3$$

$$p = \frac{1}{3}$$

$$np = 4$$

$$n = 12$$

$$\text{Thus possible.}$$

Q18 Text Solution:

Let $x = \{\text{Number of Heads Turning up}\}$

& Let Head = Success then $p = P(H) = \frac{1}{3}$ & $q = P(H) = \frac{2}{3}$

$P(3^{\text{rd}} \text{ Head in } 5^{\text{th}} \text{ toss}) = P(\text{exactly 2H in } 1^{\text{st}} 4 \text{ Tosses})$

$\times P(H \text{ is } 5^{\text{th}} \text{ toss})$

$$= \left[{}^n C_r P^r q^{n-r} \right] \times \frac{1}{3}$$

$$= \left[{}^4 C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^{4-2} \right] \times \frac{1}{3} = 6 \times \frac{2^2}{3^4} \times \frac{1}{3}$$

$$= \frac{8}{81} \text{ i.e.}$$

Q19 Text Solution:

$$X \sim B(n, p)$$

$$\text{Mean} = np$$

$$\text{Variance} = npq$$

$$\text{Variance } (x/n) = \frac{1}{n^2} \text{Var}(x)$$

$$= \frac{npq}{n^2}$$

$$\frac{1}{n} P(1 - p)$$

Q20 Text Solution:

$$P(\text{Student not having swing flu}) = \frac{9}{10}$$

$$P(4 \text{ student safe}) =$$

$${}^n C_r p^r q^{n-r} = {}^5 C_4 \left(\frac{9}{10}\right)^4 \left(\frac{1}{10}\right)^1 = 5 \times \frac{9^4}{10^5}$$

$$P(5 \text{ student safe}) = {}^5 C_5 \left(\frac{9}{10}\right)^5 \left(\frac{1}{10}\right)^0 = 1 \times \frac{9^5}{10^5}$$

$$\text{Total} = P(4 \text{ student safe}) + P(5 \text{ student safe}) =$$

$$5 \times \frac{9^4}{10^5} + \frac{9^5}{10^5} = \frac{9^4 \times 14}{10^5} = \left(\frac{9}{10}\right)^4 \times \frac{7}{5}$$

Q21 Text Solution:

$$p = 0.1, n = 400$$

$$\text{Mean} = np = 400 \times 0.1 = 40$$

$$\text{Standard deviation} = \sqrt{npq}$$

$$= \sqrt{400 \times 0.1 \times (0.9)}$$

$$= \sqrt{400 \times 0.09}$$

$$= 20 \times 0.3 = 6$$

Q22 Text Solution:

$$\text{Mean of the binomial distribution} = np = 2.5$$

$$\text{Variance of the binomial distribution} = npq = 1.875$$

$$np = 2.5 \dots (i)$$

$$npq = 1.875 \dots (ii)$$

$$\frac{npq}{np} = \frac{1.875}{2.500} = \frac{1875}{2500}$$

$$\Rightarrow q = 0.75$$

$$\text{since } p = 1 - q = 1 - 0.75 = 0.25$$

$$\text{Again since, } np = 2.5 \text{ or } n \cdot (0.25) = 2.5 \Rightarrow n = \frac{2.5}{0.25} = 10.$$

Now, with $n = 10$, $p = 0.25$ and $q = 0.75$, the binomial probability distribution is

$$p(r) = {}^{10} C_r (0.25)^r (0.75)^{10-r}, r = 0, 1, 2, \dots, 10$$



Where r is the number of successes.

Q23 Text Solution:

$$E(x) = np = 100 \times 1/6 = 16.7 \approx 17$$

So, 17 out of 100 are expected to fall 6.

$$V(x)$$

=

$$np(1-p) = 100 \times 1/6 \times (1 - 1/6) = 13.9$$

So, variance is number of 6's = 13.9.

Q24 Text Solution:

$$P(X=1) + 2P(X=0) = 12P(X=2)$$

$$\frac{e^{-\lambda} \lambda^1}{1!} + 2 \frac{e^{-\lambda} \lambda^0}{0!} = 12 \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\lambda + 2 = 6\lambda^2$$

$$= 6\lambda^2 - \lambda - 2 = 0$$

$$(3\lambda - 2)(2\lambda + 1) = 0$$

$$\lambda = \frac{2}{3}, \frac{-1}{2}$$

But λ = Variance for Poisson Distribution So it can't be -ve

Hence $\lambda = 2/3$.

$$\text{Now } P(X=0) = \frac{e^{-\lambda} \lambda^0}{0!}$$

$$= e^{-2/3}$$

$$= 0.5/34$$

Hence option (c) is correct Answer.

Q25 Text Solution:

NA

Q26 Text Solution:

NA

Q27 Text Solution:

$$P = 1\%, n = 100, \lambda = np = 1$$

$$P(K) = \frac{e^{-1}(1)^k}{k!} = \frac{e^{-1}}{k!}$$

$$P(4 \text{ or more faulty}) = P(4) + P(1) + \dots + P(100)$$

$$= 1 - [P(0) + P(1) + P(2) + P(3)] = 1 - [e^{-1}]$$

$$\left[\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} \right]$$

$$= 1 - e^{-1} \left[1 + 1 + 0.5 + \frac{1}{6} \right] = 1 - \frac{8}{3e}$$

$$= 0.019$$

Q28 Text Solution:

Mean of Poisson distribution $\lambda = np$

$$n = 10, p = \frac{1}{500}$$

$$\text{Then, } \lambda = 10 \times \frac{1}{500} = 0.02$$

$$P(x=r) = \frac{e^{-\lambda} \lambda^r}{r!}, r = 0, 1, 2, \dots$$

(i) Probability of numbers of defective blade.

$$P(r=0) = \frac{e^{-0.02}(0.02)^0}{0!} \cdot e^{-0.02} = 0.9802$$

Then number of packets with numbers of defective blade = $1,00,000 \times 0.9802 = 98020$

(ii) Probability of one defective blade

$$P(r=1) = \frac{e^{-0.02}(0.02)^1}{1!} \cdot e^{-0.02} = 0.019604$$

$$= 0.019604$$

$\therefore$ Number of packets with one defective blade = $1,00,000 \times 0.019604 = 1960$

(iii) Probability of two defective blades

$$P(r=2) = \frac{e^{-0.02}(0.02)^2}{2!} = 0.000196$$

$\therefore$ Number of two defective blades = $1,00,000 \times 0.000196$

$$= 19.6 \cong 20.$$

Q29 Text Solution:

$$\lambda = np = 500 \times \frac{1}{1000} = \frac{1}{2}$$

$$P(X=2) = \frac{e^{-\lambda} \lambda^2}{2!} = e^{-0.5} \frac{(0.5)^2}{2} = 0.07582$$

